

UrbanHeat — Simple Project Guide & Reference

Everything Explained From Scratch (For Students, Professors & Viva Preparation)

1. What Is This Project All About? (The Simple Analogy)

Imagine a giant city like **Mumbai** during summer. Standard weather apps (like Google Weather or iPhone Weather) show just **one single temperature for the entire city** (e.g. 'Mumbai is 33°C').

But anyone living in Mumbai knows that is not true! If you stand under a big green tree in **Sanjay Gandhi National Park** or **Malabar Hill**, it feels pleasant. But if you stand in **Dharavi, Kurla, or Dongri** surrounded by concrete walls, metal tin roofs, and asphalt roads, it feels like an oven!

Core Problem We Solve: Standard weather apps treat the whole city as one block. Our project, **UrbanHeat**, acts like a microscope for heat. It breaks Mumbai down into **24 municipal neighborhoods (Wards)** to show exactly where it is boiling hot and where it is cool.

2. Which Datasets Did We Take? (Where Did Data Come From?)

To build this website, we collected 3 main types of data:

Dataset Name	Where It Came From	What Information It Gives Us
1. MCGM 24 Ward Boundaries GeoJSON	Public GIS Spatial Datasets (DataMeet / MCGM BMC)	The exact latitude/longitude boundary shapes for all 24 Mumbai administrative wards (from Colaba in South Mumbai to Dahisar/Mulund in North).
2. Satellite Thermal & Infrared Data	USGS Landsat-9 Satellite & Sentinel-2 Satellites	Thermal sensors in space measure the ground skin temperature of roofs and roads, and optical cameras measure how green (plants) or concrete an area is.
3. Real-Time Live Weather API	Open-Meteo REST Meteorological API	Live hourly air temperature, humidity %, wind speed, solar radiation (W/m²), and ground soil temperature directly over the internet.

3. All Terms Required to Calculate Heat (Explained Simply)

Here are the 7 core terms you need to know to answer any question:

- **Air Temperature (T_{air}):** The regular temperature of the air around you (measured 2 meters high in the shade). Example: 33°C.
- **Land Surface Temperature (LST):** The radiative *skin temperature* of actual roads, concrete roofs, and soil! On a sunny afternoon, dark asphalt roads reach **45°C - 50°C** even when air temp is 33°C!
- **Relative Humidity (%):** The percentage of water vapor floating in the air. High humidity stops human sweat from evaporating, making heat feel unbearable.
- **Heat Index (HI):** The '**Real-Feel**' temperature combining Air Temp + Humidity. In coastal Mumbai, 34°C air + 75% humidity feels like **42°C** to the human body!

- **NDVI (Green Plant Index):** Score from -1 to +1 showing how many trees and plants exist. High NDVI (>0.35) = Lots of green trees that shade and cool the neighborhood.
- **NDBI (Concrete/Building Index):** Score from -1 to +1 showing how much concrete, asphalt, and building roofs exist. High NDBI (>0.65) = Dense concrete trapping heat.
- **HVI (Heat Vulnerability Index):** Our total combined risk score (0 to 100). Higher score = Higher danger of heat stress for citizens in that ward!

4. What Is K-Means Machine Learning & How Is It Used on Our Website?

What is K-Means? (Explained Like You're 10 Years Old):

Imagine you have a box of 24 different colored toys. You want to sort them into **3 groups** based on their color and size, but you don't know the names of the groups yet.

K-Means is a smart computer sorting algorithm (unsupervised machine learning). The '**K**' stands for the number of groups you want (we chose **K = 3**). The computer looks at all 24 Mumbai wards without any human bias, compares their ground temp (LST), concrete level (NDBI), greenery (NDVI), and population, and automatically groups them into 3 distinct microclimate types!

The 3 Microclimate Groups (Clusters) Created by K-Means:

Cluster Group	Archetype Name	Characteristics & Examples in Mumbai
Cluster 0 (Red)	Concrete Thermal Hotspot	High surface temperature (>37°C), dense concrete roofs (NDBI > 0.68), almost no trees, high population density. <i>Examples: Dharavi, Kurla, Dongri.</i>
Cluster 1 (Green)	Vegetated Thermal Buffer	Cooler surface temperature (<33.5°C), high tree canopy cover (NDVI > 0.35). <i>Examples: Sanjay Gandhi National Park, Powai, Malabar Hill.</i>
Cluster 2 (Blue)	Coastal Moderate Microclimate	Moderated by cool ocean sea breezes, moderate built density. <i>Examples: Colaba, Bandra West, Versova.</i>

How K-Means is Used in Our Website:

1. Our Python backend runs the **scikit-learn K-Means algorithm** on the dataset when the server starts.
2. Each of the 24 wards gets assigned a Cluster ID (0, 1, or 2).
3. On the interactive map on our website, when a user selects the '**Microclimate Cluster**' layer, the map automatically colors Cluster 0 Red, Cluster 1 Green, and Cluster 2 Blue!

5. What-If Urban Cooling Simulator (Interactive Feature)

We built an interactive slider tool on our website where urban planners or students can test cooling solutions:

- 'What if we plant 20% more green trees (Increase NDVI) in Dharavi?'
- 'What if we paint metal roofs white to reflect heat (Decrease NDBI)?'

Our mathematical model (Ridge Regression) instantly calculates: '**Surface temp will drop by 2.3°C and risk score will drop by 12 points!**'

6. How Everything Is Connected (Step-by-Step Architecture)

1. **Frontend (Website UI):** Built using **React 19 + TypeScript + MapLibre GL JS**. It renders the interactive map, search bar, legend, and charts.
2. **Backend (Python FastAPI):** Runs behind the scenes on port 8000. It queries Open-Meteo live weather servers, disaggregates temperature per ward, runs the K-Means algorithm, and sends clean JSON data to the website.
3. **Search & Auto-Zoom:** When a user searches 'Dharavi' or 'Bandra', the map smoothly flies and zooms straight into that ward, highlighting its boundary and thermal sampling dots!

30-Second Viva Answer for Professors:

'Sir/Ma'am, UrbanHeat disaggregates Mumbai's weather into 24 wards using Landsat-9 surface skin temp (LST) and Open-Meteo real-time atmospheric data. We use **Unsupervised K-Means Clustering (K=3)** to group wards into Concrete Hotspots, Vegetated Buffers, and Coastal Zones, and an **Empirical Ridge Regression model** to simulate temperature reduction when tree canopy or cool roofs are added.'